

How is Reliance on AI Measured? Mapping and Evaluating Measurement Approaches in Human-AI Interaction

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Abstract: Appropriate reliance on AI, following system recommendations when correct and overriding them when incorrect, is the central behavioural objective of human–AI interaction research. Despite the importance of this objective, researchers draw on a heterogeneous set of behavioural, self-report, performance-based, and process-level measures, with little principled guidance on their suitability and no shared reporting standard to enable cross-study comparison. This paper presents a systematic review of how reliance is measured across 67 HCI papers published between 2014 and 2026. We develop a taxonomy of four measurement categories and evaluate each against seven methodological dimensions grounded in measurement theory and human factors research. Sixty-one percent of included papers combine two or more measure types, yet the majority report divergent signals across instruments. We identify five structural problems limiting cumulative progress: construct ambiguity across definitions, the conflation of trust and reliance as interchangeable constructs, static measurement of an inherently dynamic construct, an ecological validity gap driven by crowdsourced single-task designs, and measurement fragmentation that prevents cross-study synthesis. We translate this diagnosis into a prescriptive measurement framework providing measure selection guidance by research question type and a minimum reporting standard applicable to all reliance studies.

Keywords: reliance measurement, human-AI interaction, appropriate reliance, over-reliance, under-reliance

1. Introduction

AI-based decision support is now embedded in consequential professional practice. Clinicians consult AI diagnostic systems for cancer screening and disease classification (Cabitza et al., 2025; Doh et al., 2025); judges and parole officers encounter algorithmic risk scores that bear on liberty (Pruss et al., 2023); financial analysts work alongside AI forecasting tools that shape portfolio decisions (He et al., 2024). In each of these settings, the critical question is not whether the AI is accurate, but whether the human appropriately acts on its output. A clinician who defers to an AI recommendation without the capacity to assess it is over-relying. Similarly, one who ignores consistently correct AI guidance because it conflicts with their intuition is under-relying. Over-reliance has been associated with diagnostic errors and unjust decisions (Goddard et al., 2014; Pruss et al., 2023); under-reliance foregoes the complementary performance gains that motivate deploying AI systems in the first place

(Bansal et al., 2019; Guo et al., 2024). Appropriate reliance, the calibrated, context-sensitive decision to follow or override AI advice, is the behavioural target that human-AI interaction research aspires to support.

The ambition of the research community to facilitate appropriate reliance is clear, what is less clear is whether the research practices currently used to study reliance are adequate to the task. It has been shown that appropriate reliance is a demanding construct: it requires ground truth to evaluate, temporal measurement to track, and ecologically valid settings to generalise (Lee & See, 2004; Parasuraman & Riley, 1997). It is also a contested construct. Researchers variously operationalise it as a binary trial-level decision, a continuous adjustment weight, a calibration state between trust and system accuracy (Naiseh et al., 2023; Naiseh et al., 2021), a habitual dispositional pattern (Holstein et al., 2025; Naiseh et al., 2024), or a volitional choice to consult AI at all (Küper et al. 2025). Each operationalisation implies a different study design, a different instrument, and a different standard for what counts as appropriate. Construct heterogeneity and measurement heterogeneity are not independent problems: the diversity of definitions in the literature has produced a corresponding diversity of instruments, and that diversity makes cross-study synthesis and comparison invalid.

Research output on human-AI reliance has grown since 2021, driven by widespread deployment of AI decision-support and introduction of LLMs in everyday tasks, intensifying concern about automation bias and inappropriate AI dependence (Buçinca et al., 2021; Wischniewski et al., 2023), and growing demands from both researchers and practitioners for actionable guidance on how to calibrate human-AI collaboration (Naiseh et al., 2024). Yet the instruments through which this research is conducted have never been systematically examined. Researchers select measures based on convenience, precedent, or task structure rather than on principled alignment between construct and operationalisation. The result is a measurement ecosystem in which studies nominally addressing the same question, does this AI design reduce over-reliance?, produce incompatible results not because the phenomenon is inconsistent, but because the instruments are not measuring the same thing. The accumulation of knowledge that would justify this volume of research has not investigated.

The absence of measurement scrutiny is not an oversight unique to this field. Studies of measurement practice in adjacent literatures, clinical decision-making (Goddard et al., 2014), automation trust (Parasuraman & Riley, 1997), and the judge-advisor paradigm in judgment and decision-making (Yaniv & Kleinberger, 2000; Harvey & Fischer, 1997), have each found that theoretical progress stalled until the field turned its attention inward and asked whether its instruments were valid, comparable, and fit for purpose. This paper conducts that investigation for AI reliance research in HCI. We present a systematic review of how reliance on AI is measured across 67 papers published between 2014 and 2026. Our object of study is not the empirical findings of these papers, what interventions increase or decrease reliance, but the instruments through which those findings are produced: how reliance is defined, operationalised, and evaluated, and whether the chosen instruments are adequate to the theoretical claims being made. We

develop a taxonomy of four measurement categories and their sub-types, evaluate each against seven methodological dimensions grounded in measurement theory and human factors research, and diagnose the structural problems in current practice that limit the field's capacity to produce cumulative, comparable, and actionable findings. We also translate these findings into a prescriptive measurement framework; a measure selection guide organised by research question type, and a minimum reporting standard applicable to all reliance studies.

The paper is organised as follows. Section 2 situates reliance within the conceptual landscape of human-AI interaction, clarifying its relationship to trust, acceptance, compliance, and automation bias. Section 3 describes the systematic review method, including search strategy, inclusion criteria, and coding procedure. Section 4 presents the taxonomy of measurement approaches, with each category evaluated against seven dimensions. Section 5 develops the cross-cutting critique of current measurement practice. Section 6 presents the prescriptive framework, draws out the theoretical implications of the corpus, and addresses limitations and future directions.

2. Background: Reliance and Its Neighbouring Constructs

2.1. Defining Reliance

Reliance, as used in this review, refers to the behavioural act of acting on AI output: accepting a recommendation, adjusting an estimate, or deferring a decision to an algorithmic system. This definition follows the behavioural tradition established by Parasuraman and Riley (1997), who distinguished reliance, the decision to use an automated system's output, from trust, which they treated as an attitudinal precursor to that decision. Lee and See (2004) formalised this distinction, defining trust as "an attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability," and reliance as the observable behavioural consequence when that attitude is acted upon. The distinction is not merely terminological: trust and reliance can dissociate empirically, a finding that recurs throughout the examined papers and motivates much of our critique in Section 5.

Two failure modes of reliance are central to the literature and to this review. Over-reliance occurs when a user follows an AI recommendation that is incorrect, deferring when they should have overridden (Naiseh et al., 2023). Under-reliance occurs when a user ignores an AI recommendation that is correct, overriding when they should have deferred (Lee & See, 2004; Buçinca et al., 2021). Appropriate reliance requires avoiding both: following the AI when it is correct and resisting it when it is wrong (Schemmer et al., 2022). This bidirectional standard is

what distinguishes reliance research from simpler compliance research. It demands that the evaluative framework be sensitive to AI correctness, not merely to whether the user followed the system.

2.2. Adjacent Constructs

Several constructs border reliance without being equivalent to it, and conflating them is a source of measurement error. **Trust** is an attitude or intention which is a psychological state that predicts, but is not identical to, behavioral reliance (Lee & See, 2004; Jian et al., 2000). A user can report high trust in an AI system and nonetheless routinely override it, as Scharowski et al. (2023) and Banovic et al. (2023) demonstrate empirically. Trust scales measure this attitude; they do not measure what users do. **Acceptance** refers to the intention to use a system, typically operationalised through technology acceptance models (Davis, 1989; Venkatesh et al., 2003). Acceptance predicts adoption at the system level but is insensitive to trial-level reliance decisions, e.g., a user who accepts an AI tool may still under-rely on any given recommendation. **Compliance** is behaviourally closer to reliance but normatively neutral: it records whether a user followed a recommendation without evaluating whether doing so was appropriate. Agreement rate, the proportion of recommendations accepted, is the standard compliance measure and its limitation is precisely that it cannot distinguish appropriate reliance from automation bias without reference to AI correctness. **Automation bias** describes the tendency to over-weight automated recommendations, producing both errors of commission (following incorrect AI output) and errors of omission (failing to catch AI errors) (Parasuraman & Manzey, 2010). It is a directionally specific failure mode, a sub-case of over-reliance, rather than a measure of reliance calibration.

2.3. Why Measurement Is Theoretically Demanding

Appropriate reliance is harder to measure than adjacent constructs for three reasons that structure the taxonomy in Section 4. First, it requires ground truth: without knowing whether the AI was correct on each trial, it is impossible to distinguish appropriate deference from automation bias (Lee and See 2004, Parasuraman and Riley 1997; Schemmer et al. 2022). Second, it requires a pre-AI baseline: to know whether a user changed their decision in response to AI advice, their initial unaided decision must be recorded, necessitating a two-stage paradigm (Yaniv and Kleinberger 2000; Schemmer et al., 2022; He et al., 2023a). Third, it is not a static state: reliance evolves as users accumulate experience with a system, receive feedback on prior decisions, and update their mental model of AI reliability (Yaniv & Kleinberger, 2000; Li et al., 2023; Naiseh et al., 2023). A single aggregate score across trials captures only a cross-sectional snapshot of a longitudinal process. These three demands (ground truth, two-stage design, and temporal resolution) are not always achievable. The five measurement

categories identified in Section 4 represent different positions in the trade-off space they define, and the evaluative framework developed in Section 4.0 makes those trade-offs explicit.

3. Method

3.1. Search Strategy

We searched three databases, Google Scholar, IEEE Xplore, and the ACM Digital Library, using search terms combining reliance, human-AI interaction, and decision support with measurement-oriented terms: *reliance measurement, over-reliance, under-reliance, trust calibration, AI decision support, appropriate reliance, and automation bias* (Check Appendix A for the full search strings). No publication date restriction was applied; the corpus spans 2014 to 2026 as a natural consequence of when the relevant literature emerged. The search was restricted to English-language publications. Duplicated contributions were removed across the three sources prior to screening. The search and screening process followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021) and is summarised in Figure 1 (PRISMA flow diagram).

3.2. Selection Process

Screening proceeded in two sequential stages, each applying progressively more restrictive criteria to the candidate pool.

- **Stage 1 - Title and abstract screening.** Following deduplication, 12,341 records were identified across the three databases. Given the volume of records, an abstract-level filtering step was applied before manual review, following established practice in high-volume systematic reviews where full manual screening of all records is not feasible (Beller et al., 2018). The purpose of this step was conservative exclusion: to remove records that were unambiguously outside scope while retaining all plausibly relevant papers for manual assessment. Specifically, abstracts were scanned to exclude three classes of record — (1) papers in which reliance on AI was not the focus of the study, appearing only as passing terminology; (2) papers with no human participant component or new proposed measure, such as purely technical AI systems papers or simulation studies; and (3) papers evaluating AI systems without reporting a behavioural measure or reliance outcome. To operationalise these exclusion criteria consistently across 12,341 records, abstracts were required to contain at least one term from each of two indicator sets (a) a human participant indicator (user study, participants, experiment, empirical study, controlled study) and (b) a reliance or

measurement indicator (measure, scale, rating, score, performance, accuracy, reliance, trust). Borderline cases, abstracts where relevance was unclear from the keyword signal alone, were retained rather than excluded and passed to manual review. To assess the sensitivity of this filtering step, a random sample of 200 records excluded at this stage was subsequently reviewed manually; the results are reported alongside the PRISMA flow diagram in Figure 1. The filtered set was then manually reviewed by the first author against the primary inclusion criterion: the paper must address human reliance on AI or algorithmic output in a decision-making context. This stage yielded 203 candidates for full-text review.

- **Stage 2 - Full-text screening.** Each paper was read in full and assessed against the primary inclusion gate: the paper must define or present a reliance measure based on our definition which is an operationalisation of how human reliance on AI output is quantified or assessed. This includes behavioural measures, self-report instruments, performance-based and calibration frameworks, process-level measures, and composite or multi-method approaches. Papers were excluded if they discussed reliance conceptually without operationalising it, if the measure presented was a replication of an existing instrument with no methodological contribution, or if the paper was not retrievable in full text. This stage yielded the final corpus of 67 papers. The screening process is summarized in Figure 1.

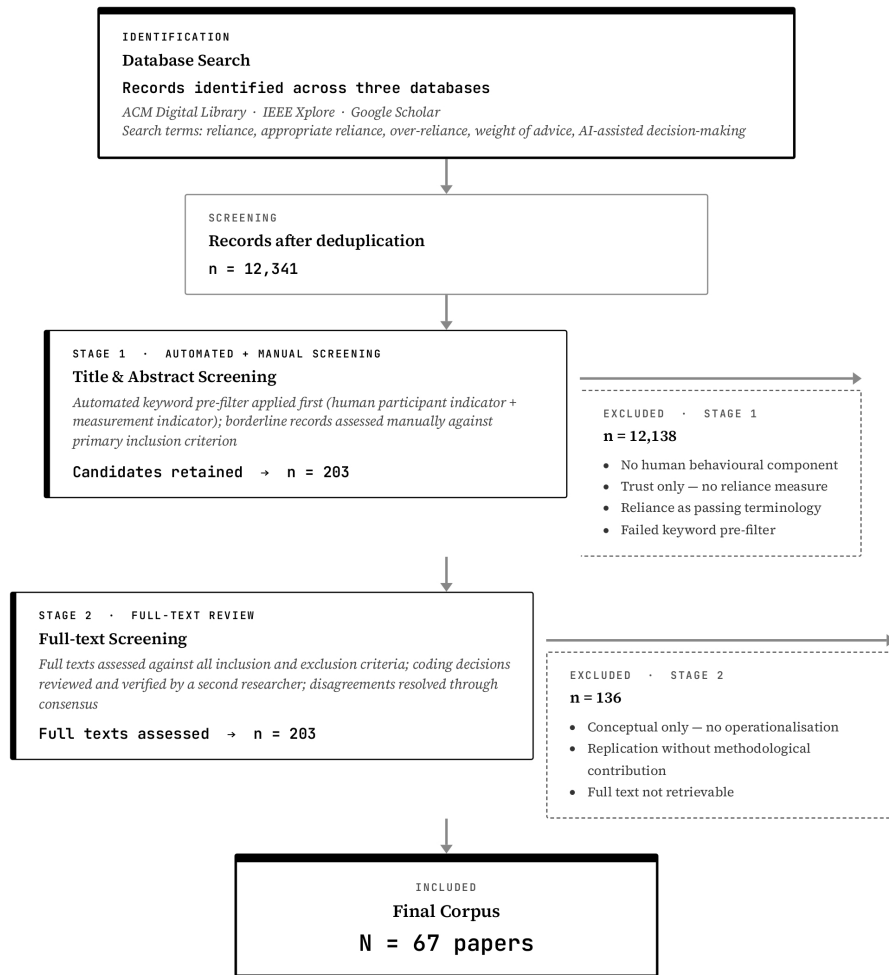


Figure 1. Two-stage screening process applied to the multi-database search. Stage 1 assessed titles and abstracts for relevance to human reliance on AI in decision-making contexts; Stage 2 applied the primary inclusion gate requiring a novel reliance measure. Final sample: N = 67 papers spanning 2014–2026.

3.3. Data Extraction and Coding

Each of the 67 included papers was coded using a structured 10-column extraction scheme covering: (1) whether reliance was explicitly defined as a construct (Y/N); (2) the verbatim definition where provided; (3) the primary measure type (behavioral, performance-based and calibration, self-report, or process-level); (4) the specific measure or instrument used; (5) whether trust calibration was used as a proxy for reliance (Y/N); (6) the application domain; (7) the task type; (8) the study setting (lab, online, field); (9) sample size; and (10) whether ground truth was available for each trial (Y / Partial / N). The four measure type categories were derived inductively from full-text reading of the measure descriptions: no category scheme was imposed a priori.

Coding was conducted by the first author, who read each paper's methods section in full, identified the specific instrument or operationalisation used, and assigned a primary measure type label. Where a paper used instruments from more than one category, the component types were recorded separately and the co-occurrence structure was documented rather than collapsed into a composite code. The second and third authors

independently reviewed all category assignments against the original papers; disagreements were discussed and resolved through consensus, resulting in revisions to a subset of assignments before final coding was agreed across all three authors. The coding scheme, all category assignments, ambiguous-case coding rules, and a quick reference for all key reliance measures cited in the corpus are documented in Appendix B. The full coded dataset is publicly available at https://osf.io/f5bt9/overview?view_only=a3627a2301cc42cbb44dd0d5f1e99297

4. Results

This section presents the taxonomy of reliance measurement approaches identified across the 67 included papers, spanning publications from 2014 to 2026 with a pronounced concentration in the most recent three years. Inductive coding of the full-text measurement descriptions yielded four broad categories: behavioral measures (n=40), performance-based and calibration measures (n=24), self-report measures (n=24), and process-level measures (n=11). These categories are not mutually exclusive: papers using instruments from two or more categories simultaneously account for 41 of the 67 included studies (61%), a figure that reflects both the perceived inadequacy of any single measure and the practical availability of multiple instruments within a controlled experimental design. The co-occurrence structure across all four categories is shown in Table 1. Although reliance is explicitly defined as a construct in 52 of the 67 papers (78%), the remaining 15 papers (22%) operationalize a reliance measure without providing a formal conceptual definition; a pattern that itself reveals something about the field's relationship to its central construct.

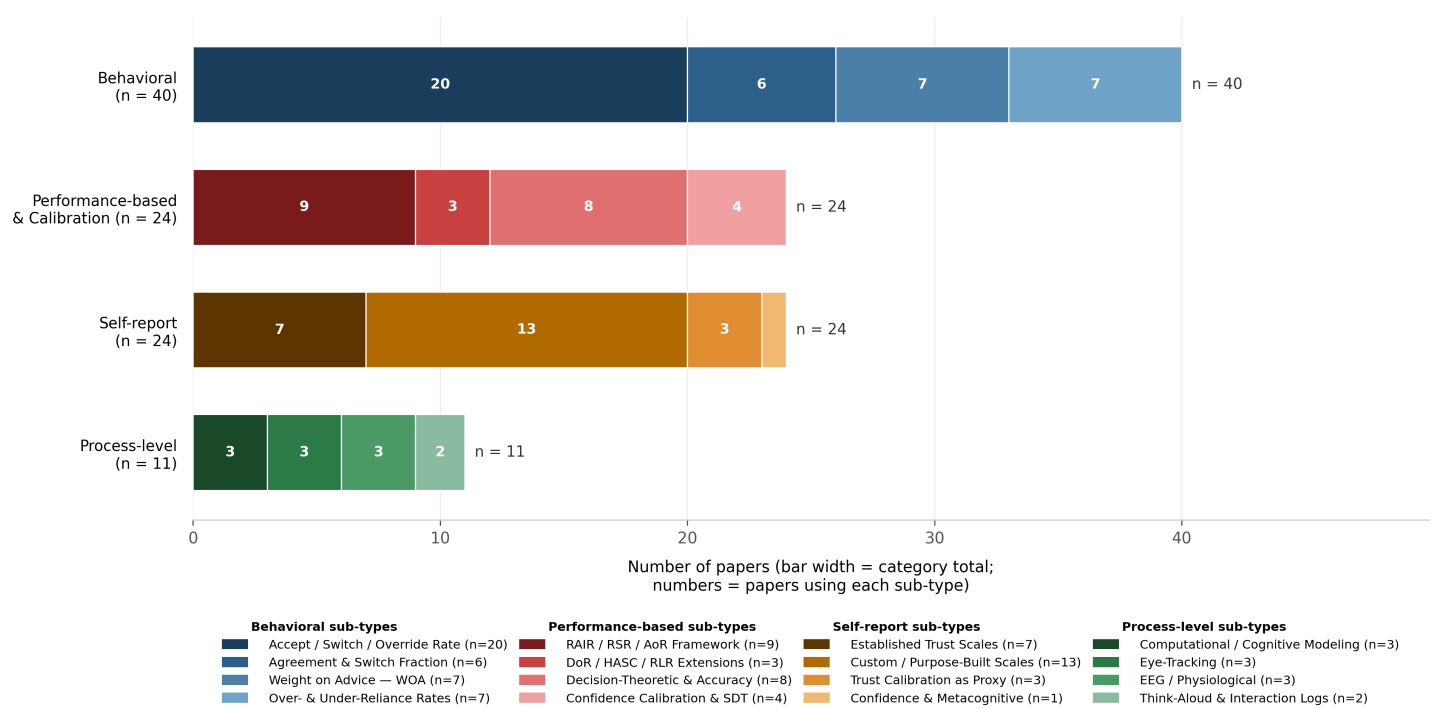


Figure 2. Distribution of included papers across the five reliance measure categories and their sub-types (N = 67). Each bar represents one category; segments within each bar indicate the number of papers using each sub-type. Papers deploying instruments from more than one category are counted in each relevant category; totals therefore exceed N = 67.

Table 1. Co-occurrence of Measure Types Across the 67 Included Papers

Combination	n	Example Papers
Single measure type		Subtotal: n = 25
Behavioural only	10	He et al. (2023); Pruss (2023); Zhang et al. (2020)
Performance-based & Calibration only	9	Schemmer et al. (2023); Guo et al. (2024); Cabitza et al. (2025)
Self-report only	3	Leist et al. (2025); Hou et al. (2025)
Process-level only	3	Li et al. (2023); Lu et al. (2024); Tejeda et al. (2022)
Two measure types combined		Subtotal: n = 29
Behavioural + Performance-based & Calibration (B + P)	9	Bo et al. (2025); He et al. (2024); Spatola (2024)
Behavioural + Self-report (B + S)	13	Papenmeier et al. (2022); Kahr et al. (2024); Kim et al. (2025)
Behavioural + Process-level (B + Pr)	3	Zhou et al. (2025); Pearson et al. (2026)
Performance-based & Calibration + Self-report (P + S)	2	Steyvers et al. (2024); Dogru & Krämer (2025)
Performance-based & Calibration + Process-level (P + Pr)	2	Zhang et al., (2024); Chen et al. (2023)
Self-report + Process-level (S + Pr)	0	—
Three measure types combined		Subtotal: n = 11
Behavioural + Performance-based + Self-report (B + P + S)	7	Mehrotra et al. (2024b); Salimzadeh et al. (2024); Küper et al. (2025)
Behavioural + Performance-based + Process-level (B + P + Pr)	0	—
Behavioural + Self-report + Process-level (B + S + Pr)	4	Jui et al. (2025); Cao & Huang (2022); Mei et al. (2025)
Performance-based + Self-report + Process-level (P + S + Pr)	0	—
All four measure types		Subtotal: n = 2
Behavioral + Performance-based + Self-report + Process-level (B + P + S + Pr)	2	Wischnewski et al. (2023); Lee & Chew (2023)
Total papers in corpus	67	Papers using multiple measure types: 41 (61%)

Note. B = Behavioral; P = Performance-based & Calibration; S = Self-report; Pr = Process-level. Papers are counted once only in their exact combination. Category totals in Figure 2a exceed 67 because papers using multiple measure types are counted in each applicable category.

To support systematic comparison across the four categories and their sub-types, we evaluate each measure along six dimensions that capture the key methodological and conceptual trade-offs in reliance measurement. These dimensions are not novel: each is grounded in established criteria from measurement theory, human factors research, and the broader literature on human-AI decision-making, and together they constitute the evaluative lens through which Sections 4.1 - 4.4 are written. They are defined as follows and summary of the measures evaluation can be found in Table 2.

- **Ground truth requirement.** A measure requires ground truth when it can only be interpreted, and in particular, when it can only distinguish appropriate from inappropriate reliance, by reference to the objectively correct answer on each trial. The necessity of ground truth for normative reliance assessment has been argued by Lee and See (2004), Parasuraman and Riley (1997), and most recently formalized by Schemmer et al. (2022), who demonstrated that behavioral measures without ground truth conditioning can only index the direction of AI influence rather than its appropriateness. Studies without ground truth can measure compliance but not calibration.
- **Two-stage design requirement.** A two-stage paradigm, in which participants record an initial decision before the AI recommendation is revealed, and a final decision afterward, is a structural prerequisite for most

behavioral and performance-based measures, because it is only by observing the pre-AI baseline that the influence of AI advice can be isolated. The judge-advisor paradigm formalised by Yaniv and Kleinberger (2000) and subsequently adopted in human-AI reliance research (Schemmer et al., 2023; He et al., 2023b) is the canonical implementation. Measures that do not require this structure, most self-report instruments and some process-level approaches, trade experimental control for ecological validity, since real deployed systems rarely impose a forced pre-AI response before revealing recommendations.

- **Sensitivity to partial deference.** Binary accept/reject measures treat any final decision that does not fully adopt the AI recommendation as non-reliance, failing to capture cases where participants adjust toward the AI without fully adopting it. This limitation was identified early in the judge-advisor literature (Harvey & Fischer, 1997; Yaniv & Kleinberger, 2000) and is addressed by continuous measures such as Weight on Advice (WOA; Scharowski et al., 2023) and process-level instruments such as eye-tracking (Cao et al., 2022) and cognitive modeling (Tejeda et al., 2022). In tasks with continuous response scales, medical severity scoring, price estimation, probabilistic judgment, sensitivity to partial deference is essential; in binary classification tasks it is less critical.
- **Separation of over- and under-reliance.** Appropriate reliance requires both that participants follow the AI when it is correct (avoiding under-reliance) and that they resist it when it is incorrect (avoiding over-reliance) (Naiseh et al., 2023; Buçinca et al., 2021). A measure that collapses these two behaviorally distinct patterns into a single aggregate score, such as a raw agreement rate or a global accuracy metric, cannot identify which failure mode is driving suboptimal human-AI team performance. The RAIR/RSR framework (Schemmer et al., 2022, 2023) was specifically designed to separate the two, and the empirical importance of doing so is demonstrated by several multi-method studies that find RAIR and RSR dissociating under the same experimental manipulation (Morrison et al., 2024; He et al., 2023; Bo et al., 2025).
- **Normative grounding.** A measure has normative grounding when it evaluates reliance behavior against an external standard of what constitutes appropriate reliance; that is, when it can distinguish not just whether the participant followed the AI but whether doing so was warranted. Normative grounding requires, at minimum, ground truth availability, and in its strongest form requires an explicit benchmark such as the rational baseline proposed by Guo et al. (2024) or the Appropriateness of Reliance score proposed by Schemmer et al. (2023). Self-report trust scales have low normative grounding because they measure the participant's attitude rather than whether that attitude was epistemically justified. The concept of normative grounding is closely related to the notion of trust calibration (Lee & See, 2004; Wischniewski et al., 2023), but is broader: a well-calibrated trust attitude does not guarantee appropriate reliance behavior, as multiple studies in the corpus demonstrate.

- **Ecological validity.** Ecological validity refers to the degree to which the measurement context ,task, stimuli, participant population, and AI system, resembles the real-world settings in which AI reliance occurs (Brewer, 2000). It is rated Low in this taxonomy where studies rely exclusively on crowdsourced online participants performing artificial classification tasks with experimentally controlled AI accuracy levels; Medium where some naturalistic elements are present (realistic stimuli, expert participants, or real AI systems); and High where field or clinical deployment settings are used. As noted by Goddard et al. (2014) and Küper et al. (2025), the ecological validity of reliance research has important implications for the generalizability of findings: over-reliance rates observed in artificial laboratory tasks may substantially overestimate or underestimate those in real deployment contexts.
- **Practical accessibility.** This dimension captures the combined methodological burden of a measure (the equipment, design constraints, sample size requirements, and analytic expertise needed to implement it). High practical accessibility means the measure can be collected with standard online survey tools and analyzed with common statistical methods; Low accessibility means it requires specialized equipment (EEG, eye-tracking), large samples, or computational modeling expertise. Practical accessibility shapes which measures researchers reach for in practice, and partly explains the dominance of simple behavioral and self-report measures in the corpus despite the theoretical superiority of more demanding approaches (Schemmer et al., 2022; Tejeda et al., 2022).

The six dimensions are not fully independent, ground truth requirement and normative grounding are closely related, as are two-stage design and sensitivity to partial deference, but they capture distinct enough aspects of the measurement problem to justify separate treatment. Table 2 provides a summary profile of all four categories across these dimensions; the subsections that follow develop the argument for each category in detail.

Table 2. Reliance Measure Category Profiles Across Key Evaluation Dimensions

Dimension	Behavioural	Performance-based	Self-report	Process-level	Composite / Multi-method
Requires ground truth	Yes	Yes	No	Partial	Varies
Requires 2-stage design	Yes	Yes	No	Partial	Varies
Captures partial deference	No (WOA only)	No	Partial	Yes	Varies
Separates over- vs. under-reliance	Partial	Yes	No	Partial	Varies
Normative grounding	Partial	High	Low	Partial	Varies
Ecological validity	Low–Med	Low	Med–High	Med–High	Varies
Practical accessibility	High	Med	High	Low	Med
Overall measurement strength	Medium	High	Low–Medium	Medium	Medium–High
Best suited for	Detecting direction and frequency of AI influence across conditions	Evaluating whether reliance was appropriate, separating over- from under-reliance	Measuring attitudinal correlates, dispositional reliance, or naturalistic AI use	Understanding cognitive mechanisms underlying reliance decisions	Capturing reliance across behavioral, attitudinal, and normative dimensions simultaneously

Key design constraint	Two-stage paradigm; ground truth needed	50/50 AI correct/incorrect split; two-stage paradigm	Attitude ≠ behaviour; systematic over- estimation	Specialist equipment or expertise required	Participant burden; analytic complexity
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Note. "Partial" indicates the criterion is satisfied only under specific conditions. "Varies" indicates the composite category inherits the properties of its component measure types. Ecological validity is rated Low where studies rely exclusively on crowdsourced online participants with artificial stimuli; Med where some naturalistic elements are present; High where field or realistic deployment settings are used. Overall measurement strength is an evaluative synthesis across all seven dimensions. Supporting evidence for each rating is developed in Sections 4.1 - 4.5.

4.1. Behavioural Measures

Behavioural measures are the most frequent approach in the sample papers, appearing in 40 of 67 papers (60%). Their appeal is direct: if reliance is ultimately a behavioural construct, measuring what the user *does* rather than what they report offers the most unmediated empirical window onto it. Before describing the sub-types, a definitional boundary requires clarification. Behavioural measures, as used in this taxonomy, record the *direction* of the human's response to AI, whether they follow, switch, or resist, without making a normative claim about whether that response was warranted. They are distinguished from performance-based and calibration measures (Section 4.2) on a single criterion: performance-based measures evaluate reliance *against an external standard of correctness*, asking not merely what the user did but whether doing so was appropriate given what the AI actually got right or wrong. A pure behavioural measure can be computed without ground truth; a performance-based measure cannot. This boundary is theoretically clean but empirically porous: some behavioral sub-types, most notably over- and under-reliance rates, require ground truth to compute, and therefore sit at the intersection of the two categories. The rationale for retaining them within the behavioral category, rather than relocating them to Section 4.2, is that they preserve the behavioral framing: they express what the user did on AI-correct and AI-incorrect trials respectively, without converting those actions into a calibration score or comparing them to an optimal benchmark. They describe a pattern of behavior; they do not evaluate it. Where the distinction matters for interpretation, it is flagged explicitly. Behavioral measures span four recognizable sub-types that vary substantially in sensitivity and design demands.

4.1.1. Accept/Switch/Override Rate

The most elementary behavioural measure records whether, on a given trial, a participant accepts or rejects the AI's recommendation. This binary operationalisation appears in several forms, open rate and submit rate (Tolmeijer et al., 2021), agreement rate and switch rate (Naiseh et al., 2022), switch-to-incorrect rate (Cao et al., 2024) and its appeal lies in computational simplicity. Its core limitation is equally simple: without reference to whether the AI was correct on each trial, it cannot distinguish appropriate from inappropriate reliance (Lee & See, 2004). A user who accepts every recommendation scores identically whether the AI is right or wrong, conflating two normatively opposite events.

4.1.2. Over-Reliance and Under-Reliance Rates

A natural refinement conditions acceptance on AI correctness, yielding separate over-reliance and under-reliance rates. Over-reliance is the proportion of AI-incorrect trials on which the participant accepts; under-reliance is the

proportion of AI-correct trials on which the participant rejects (Vasconcelos et al., 2023; Li et al., 2025; Doh et al., 2025). Over-reliance rate is the most commonly reported sub-variant in the corpus, appearing across healthcare (Doh et al., 2025; Kim et al., 2025), coding assistance (Pitts et al., 2025), and XAI evaluation (Li et al., 2025). Goddard et al. (2014) labeled the analogous pattern "negative consultation" and documented it in 5.2% of cases among UK general practitioners. Both rates require ground truth and per-trial AI correctness; a property they share with the performance-based measures in Section 4.2. The boundary between the two categories is worth making explicit here. Over- and under-reliance rates are retained in the behavioral category because they express a frequency of behavioral events (following on AI-incorrect trials; resisting on AI-correct trials) rather than a calibration ratio or appropriateness score. They tell you how often an error pattern occurred; they do not tell you how far the participant deviated from optimal behavior or whether their overall reliance trajectory was appropriate. RAIR and RSR (Section 4.2) are the calibration counterparts: they are proportions that explicitly quantify *appropriate* following and resistance respectively, and are designed to be interpreted against a normative benchmark. The distinction, in short, is between describing a behavioral pattern and evaluating it. Both sub-types also remain insensitive to *partial* deference: in continuous response domains, medical severity scoring (Ramesh et al., 2020), price estimation (Holstein et al., 2025); binary coding represents meaningful adjustment as non-reliance.

4.1.3. *Agreement Fraction, Switch Fraction, and the He et al. Framework*

Unlike the acceptance-based measures in Section 4.1.1, which record the outcome of a single post-AI decision, Agreement Fraction and Switch Fraction are inherently comparative: both require a pre-AI baseline decision, making them structurally dependent on a two-stage paradigm and therefore sensitive to the influence of AI advice rather than merely to its adoption. He et al. (2023a) decomposed behavioural reliance into Agreement Fraction (proportion of final decisions aligning with the AI), Switch Fraction (proportion of initial-disagreement trials on which the participant switches to match the AI), and correctness-conditioned variants RAIR and RSR (Section 4.2). The decomposition matters because the two surface measures can dissociate: He et al. (2023b) found calibration interventions shifted Switch Fraction without significantly altering Agreement Fraction, a finding invisible to a single acceptance rate. The framework has been adopted across He et al. (2024, 2025), Lu et al. (2021), and Ma et al. (2024, 2025).

4.1.4. *Weight on Advice (WOA)*

Weight on Advice (Yaniv & Kleinberger, 2000) is computed as $(T2 - T1) / (AI - T1)$, where T1 is the initial estimate, T2 the final, and AI the recommendation. WOA = 0 indicates no adjustment; WOA = 1 indicates full adoption; intermediate values capture partial deference. The measure appears in seven papers (Ahn et al., 2024; Kahr et al.,

2024; Scharowski et al., 2023; Ramesh et al., 2020; Bo et al., 2025; Küper et al., 2025; Holstein et al., 2025) and is particularly suited to continuous response domains. Küper et al. (2025) found trust explained up to 24.1% of WOA variance in dermatologists; Holstein et al. (2025) showed median WOA collapsing near zero at high uncertainty, a distributional pattern obscured by mean WOA alone. WOA carries three limitations: it is undefined when AI advice equals the initial estimate; it can exceed (0,1) when participants over-adjust (requiring capping); and it assumes a continuous outcome space that does not transfer naturally to categorical tasks, e.g., Ahn et al. (2024) apply it to binary prediction and label it "behavioral trust," arguably conflating constructs.

4.1.5. Consultation Rates and Task-Specific Variants

A smaller subset operationalises reliance as the frequency of choosing to consult the AI at all. Spatola (2024) counted AI accesses per participant; Mei et al. (2025) tracked whether participants viewed AI decisions (N=7,600); Nourani et al. (2021) used interaction logs where no per-trial paradigm was feasible. This captures volitional engagement but cannot assess appropriateness. Beyond these sub-types, fifteen behavioural papers deploy task-specific variants, ConcurN/ConversionN in chess (Banovic et al., 2023), spatial user-AI agreement (Cao et al., 2022), qualitative non-use among judges (Pruss et al., 2023), SDT decomposition (Pearson et al., 2026) reflecting a task-coupled rather than construct-driven measurement culture.

4.1.6. Measurement Properties

Overall, our analysis shows that behavioural measures score well on practical accessibility, all sub-types can be implemented with standard experimental software and analysed with common statistical methods, and on ground truth availability, present in 35 of 40 papers (88%). They universally require a two-stage paradigm, which enables measurement precision but constrains ecological validity: the forced pre-AI response imposes a procedural structure absent from most real deployment contexts. Normative grounding is partial and sub-type-dependent: raw acceptance rates index compliance rather than appropriateness, while over- and under-reliance rates and the He et al. framework partially separate the two failure modes by conditioning on AI correctness. Only WOA captures partial deference; binary measures treat any non-full adoption as non-reliance. The full dimensional profile of the category is summarised in Table 2.

4.2. Performance-Based and Calibration Measures

Performance-based and calibration measures appear in 24 of 67 papers (36%) and represent the theoretically most mature category. Unlike behavioural measures that record what a participant does, this category evaluates whether the reliance behaviour was *warranted*, i.e., whether the participant followed the AI when it was correct and resisted it when it was wrong. This normative orientation makes it the closest operationalisation to the construct of *appropriate* reliance (Lee & See, 2004; Parasuraman & Riley, 1997).

4.2.1. *The RAIR/RSR Framework*

The dominant measure in the category, appearing in nine papers, is the Appropriateness of Reliance (AoR) framework (Schemmer et al., 2022, 2023). The framework requires a three-stage paradigm: participants make an initial unaided decision (Stage 1), observe the AI recommendation (Stage 2), and produce a final decision (Stage 3). Relative AI Reliance (RAIR) is the proportion of trials where the human was initially wrong and the AI correct and the participant switched to match the AI, i.e., correct AI-following relative to all trials where following was warranted. Relative Self-Reliance (RSR) is the proportion of trials where the human was initially correct and the AI wrong and the participant held firm, correct self-reliance relative to all trials where resistance was warranted. Appropriate reliance is jointly high RAIR and high RSR; the two scores can be visualised as coordinates in a two-dimensional reliance space (Morrison et al., 2024; Spitzer et al., 2025). The framework has been replicated across review classification (Schemmer et al., 2022, 2023), bird species identification (Schemmer et al., 2023; Zhang et al., 2024), logical reasoning (He et al., 2023), and financial decision tasks (He et al., 2024, 2025), becoming the field's closest approximation to a measurement standard. A key design constraint is that AI advice must be revealed after an initial decision, and the stimulus set should include a sufficient number of both AI-correct and AI-incorrect trials, e.g., Schemmer et al. (2023) recommend a 50/50 split. This introduces a potential validity concern: real deployed systems do not present 50% erroneous advice, and participants exposed to unusually high error rates may calibrate downward in ways that do not generalise to deployment (Küper et al., 2025).

4.2.2. *Extensions: DoR, the HASC Matrix, and the Bo et al. Framework*

Three papers extend the two-dimensional RAIR/RSR space. Morrison et al. (2024) represented each participant as a point in RAIR-RSR space and defined Deception of Reliance (DoR) as Euclidean distance from the ideal point (RAIR=1, RSR=1), finding that AI explanations improved RAIR without changing RSR, a dissociation invisible to a single accuracy measure. Spitzer et al. (2025) formalized DoR as the RAIR or RSR difference between correct and

incorrect AI trials, providing a direct sensitivity test for whether participants distinguish reliable from unreliable outputs. Doh et al. (2025) proposed the HASC (Human-AI System Classification) Matrix, an eight-cell framework that extends RAIR/RSR logic to three response categories and distinguishes outcome patterns, including cases where both human and AI are wrong, that the two-class framework collapses. It provides richer diagnostic information at the cost of requiring larger samples to populate all cells. Bo et al. (2025) proposed a structurally parallel framework for LLM tasks: Relative LLM Reliance (RLR), Relative Self-Reliance (RSR), and Appropriate Reliance Rate (ARR = a composite scalar). The key difference from Schemmer et al. is that denominators are defined over optimal-action sets rather than AI-correct subsets (a consequential difference when AI accuracy varies within a study). Applied to LSAT reasoning and numerical estimation with GPT-4, the framework revealed that explanation-free LLM outputs produced lower ARR largely through RSR deficits.

4.2.3. Decision-Theoretic and Accuracy-Based Approaches

Guo et al. (2024) proposed a decision-theoretic framework that defines a rational baseline R_0 (the better of human-alone or AI-alone) and a rational benchmark R^* (theoretically optimal complementary performance), evaluating human-AI teams relative to both bounds. Applied retrospectively to three existing datasets, it demonstrated that studies reporting "improved accuracy" with AI did not necessarily reflect appropriate reliance when the improvement fell below R^* . The framework is primarily diagnostic as it requires independent trial-level estimates of human and AI accuracy that are not always collected. Six papers operationalise reliance through accuracy change before and after AI exposure. Cabitza et al. (2025) compared diagnostic accuracy on vertebral fracture detection before and after AI disclosure as an automation bias index. Lee and Chew (2023) measured F1-score improvement across explanation conditions. Küper et al. (2025) found accuracy improved by only 1% with AI assistance in dermatologists despite substantial behavioral adjustment - illustrating the ceiling effects and domain knowledge that suppress AI benefit in expert populations. Accuracy-based approaches have ecological face validity but conflate appropriate and inappropriate reliance: high team accuracy with a high-accuracy AI tells us nothing about how a participant would behave if the AI's accuracy drops (Guo et al., 2024).

4.2.4. Confidence Calibration and Signal Detection Theory

Confidence calibration and signal detection theory (SDT) approaches occupy an important boundary in the taxonomy: they use subjective confidence ratings as their raw material, participants rate how certain they are in their decision before and after seeing the AI recommendation, but evaluate those ratings against ground truth, which is what places them in the performance-based category rather than self-report. The normative question is not how confident a participant feels about the AI in general, but whether their confidence tracks the AI's actual reliability trial by trial: does confidence rise when the AI is correct and fall when it is wrong? Goddard et al. (2014)

demonstrated this logic by showing that pre-advice decision confidence was the strongest single predictor of switching ($z=-3.61$, $p<.005$). Participants with low initial confidence were more likely to follow the AI regardless of whether it was correct, revealing a confidence-driven vulnerability to over-reliance. Cao et al. (2024) operationalised Confidence Change (post-AI minus pre-AI confidence) as a continuous reliance index, finding it sensitive to uncertainty presentation format in ways that binary accept/reject rates were not: the same behavioural acceptance rate could reflect very different degrees of genuine updating depending on how much confidence shifted. Pearson et al. (2026) applied full SDT decomposition, d' (discriminability) and criterion c (response bias), separating whether participants could distinguish reliable from unreliable AI outputs from whether they were globally biased toward acceptance. This psychophysical refinement is the most analytically precise instrument in the sub-type, distinguishing two qualitatively different failure modes, poor discrimination and miscalibrated bias, that aggregate behavioural measures collapse into one. The shared methodological requirement is a sufficient number of both AI-correct and AI-incorrect trials to estimate sensitivity, and approximately Gaussian signal distributions which is condition that many reliance paradigms are not designed to satisfy.

4.2.5. Measurement Properties

The category is unified by normative orientation: not just whether participants followed the AI, but whether doing so was warranted. Ground truth is available in 22 of 24 papers (92%), the highest rate across all categories, and RAIR/RSR explicitly separates over- from under-reliance which are properties that place this category at the high end of normative grounding. Three cross-cutting limitations apply: all measures require a two-stage paradigm that may not map to real-world settings where AI advice arrives before or alongside the decision; they are insensitive to the process by which reliance occurs (two participants with identical RAIR/RSR scores may have arrived there through entirely different cognitive routes); and the proliferation of related but non-identical frameworks (RAIR/RSR, DoR, HASC, RLR/ARR, decision-theoretic) creates fragmentation that prevents direct cross-study comparison (a problem addressed in Section 5). The full dimensional profile of the category is summarised in Table 2.

4.3. Self-Report and Trust Calibration Measures

Self-report measures appear in 24 of 67 papers (36%) and encompass validated trust scales, custom reliance ratings, purpose-built questionnaires, and confidence calibration metrics. What unites them is methodological accessibility: no ground truth, no two-stage paradigm, no interaction log required. What makes the category

contested is that a substantial portion uses trust calibration as a proxy for reliance (a substitution whose consequences are taken up in Section 5).

4.3.1. Established Trust Scales

The most common instruments are validated trust scales not designed to measure reliance directly: the Trust Perception Scale (TPA) (Jian et al., 2000), the cognitive-based trust scale (Madsen & Gregor, 2000), and the Trust-in-Automation (TiA) questionnaire. These function as attitudinal correlates or antecedents of reliance rather than measures of it. When deployed alongside behavioral reliance outcomes, they frequently dissociate. Scharowski et al. (2023) administered the TPA alongside WOA in a price estimation task and found no significant difference in TPA scores across conditions despite meaningful WOA variation, explicitly arguing against conflating the two. Kahr et al. (2024) administered a per-trial trust item after each criminal sentencing case, tracking its co-evolution with WOA and finding reciprocal longitudinal influence, but not reducibility. Tolmeijer et al. (2021) found self-reported trust tracked accuracy sequences more faithfully than behavioral reliance did, the opposite dissociation. Dogru et al. (2025) found trust predicted RAIR but not RSR, illustrating selective rather than uniform influence. The core limitation is instrument mismatch: these scales were designed to measure trust perception (Jian et al., 2000) or computer system trust broadly (Madsen & Gregor, 2000), not AI decision-support reliance specifically. The trust-reliance dissociation observed throughout the corpus is partly an artefact of applying the wrong instrument.

4.3.2. Trust Calibration Used as a Proxy for Reliance

Nine papers use trust calibration, the alignment between self-reported trust and the AI's actual accuracy, as a proxy for reliance without collecting a behavioural outcome. Steyvers et al. (2024) asked participants to estimate the probability that an LLM answer was correct (human confidence = 0–1) and computed Expected Calibration Error (ECE) and AUC for discrimination as the primary reliance outcomes. This psychometric approach is continuous, requires no binary accept/reject paradigm, and can be computed whenever ground truth is available, but it measures confidence in the AI rather than the decision to follow it. A participant who accurately distinguishes correct from incorrect LLM outputs but consistently ignores that knowledge behaviorally scores well on ECE while exhibiting severe under-reliance. At the other hand, Leist et al. (2025) used SUS scores, trust ratings, and interview coding to infer likely reliance in a clinical ophthalmology setting, with no behavioural measure collected at all which is a design forced by clinical feasibility but one whose central claims about reliance rest on stated intention rather than observed action. Wang et al. (2024) compared a per-trial self-reported reliance slider against behavioural agreement rate directly, finding self-reported reliance was systematically higher (participants over-reported their AI-following relative to their actual decisions).

4.3.3. Single-Item, Custom, and Purpose-Built Scales

Several papers use bespoke self-report measures targeting reliance rather than trust directly. Mei et al. (2025) included a Perceived AI Reliance item alongside behavioural and process-level measures in a pre-registered study (N=810). Cao et al. (2022) used a single Likert item alongside behavioural and eye-tracking data, finding moderate but imperfect alignment between self-reported and objective reliance(a within-paper validation rare in the sample). Pitts et al. (2025) included an Overreliance Self-Assessment alongside their behavioural over-reliance rate and found that participants systematically underestimated their own over-reliance which suggests self-report and behavioural measures can diverge not just in magnitude but in direction. The most substantive development in this sub-type is the GenAI Reliance Scale (Hou et al., 2025a, 2025b): a 16-item, 5-factor Likert instrument with factors for Reflective Use, Cautious Use, Thoughtless Use (passive acceptance; $\alpha=.72$), and Compulsive Use ($\alpha=.82$), validated via confirmatory factor analysis in educational settings. The scale operationalises reliance as a habitual dispositional pattern rather than a per-trial event, making it suited to naturalistic and longitudinal research where controlled paradigms are impractical. Its limitation is the inverse: without ground truth conditioning, it cannot distinguish over- from under-reliance, and its subscales approximate normative stance (Thoughtless Use $\approx$ over-reliance; Cautious Use $\approx$ under-reliance) rather than measuring it directly.

4.3.4. Confidence and Metacognitive Reliance

Confidence measures sit at the boundary between self-report and performance-based categories. Kim et al. (2025) found that lower pre-AI confidence predicted AI-following even when the AI was wrong. Cao et al. (2024) used Confidence Change (post minus pre) as a reliance index sensitive to uncertainty format. Küper et al. (2025) used a per-trial confidence slider as the input to WOA computation, making confidence both operationalisation and measure. This boundary ambiguity reflects a deeper conceptual question: whether reliance is fundamentally a metacognitive state (how confident am I in the AI?) or a behavioural disposition (how often do I act on it?) which is a question the selected papers has not resolved.

4.3.5. Measurement Properties

Three patterns characterize the 24 self-report papers. First, the most prevalent use is measuring trust as an antecedent or correlate of reliance, a role established scales are designed for, but which does not constitute reliance measurement. Second, in 20 of 67 papers trust calibration substitutes for behavioural measurement despite repeated empirical challenges to that substitution. Third, ground truth is absent in most self-report papers, only 10 of 24 report availability, reflecting their deployment precisely where behavioral measurement is infeasible.

This practical accessibility is genuine, but without ground truth it is impossible to determine whether self-reported reliance reflects appropriate or inappropriate AI dependence, and the category's normative grounding is correspondingly low. The full dimensional profile of the category is summarized in Table 2.

4.4. Process-Level Measures

Process-level measures appear in 11 of 67 papers (16%) and probe *how* a reliance decision is reached, i.e., the cognitive, attentional, temporal, and computational mechanisms that mediate human-AI interaction. The category is defined by its object of measurement: not what the user decided (behavioral measures), not whether that decision was warranted (performance-based measures), not what the user believed or reported (self-report measures), but the underlying processes through which a recommendation is received, evaluated, and acted upon. This distinction is important because the same observable reliance behavior can arise from very different processes: automatic deference, deliberate agreement, or attentional overload. Process-level measures are the only instruments capable of distinguishing these pathways, which is precisely why they are indispensable for theoretical accounts of reliance but rarely used in applied evaluation contexts where outcomes rather than mechanisms are the primary concern.

The sub-types within this category are heterogeneous in method but unified in purpose. Computational and cognitive models make the underlying decision process explicit through formal parameterisation. Eye-tracking and physiological measures such as EEG and HRV provide continuous, non-intrusive windows into attentional and cognitive states during interaction. Think-aloud protocols and interaction logs capture the temporal unfolding of deliberation through verbal or behavioral traces. What unites them is that they index something *prior to and generative of* the accept/reject outcome rather than the outcome itself. The cost is methodological complexity; most instruments in this category appear in only one or two papers in the corpus.

4.4.1. Computational Modelling

Li et al. (2023) fitted a Hidden Markov Model (HMM) to behavioral data from an income prediction task, treating trust as a latent state that evolves across trials and mediates accept/reject outcomes. Three model components, Trust, Trust Dynamics, and Decision Model, jointly predicted individual reliance trajectories and revealed asymmetric trust dynamics: negative experiences degraded trust more rapidly than positive experiences restored it. This asymmetry would be invisible to any cross-sectional or aggregate measure.

Lu et al. (2024) proposed a Bayesian mixture model (Mix and Match, M&M) representing the participant population as K latent decision-maker types, each with parameters for risk aversion (β) and sensitivity (δ). Applied retrospectively to three existing datasets, M&M recovered distinct reliance profiles, conservative, neutral, and

liberal AI-followers, obscured by aggregate agreement rates. Tejeda et al. (2022) applied hierarchical Bayesian modeling to a concurrent advice paradigm (AI shown simultaneously with the task rather than sequentially), estimating per-participant switching probabilities as a function of confidence, accuracy condition, and trial history. Their model is one of the few process-level tools suited to ecologically realistic settings where AI recommendations arrive before or alongside the human decision; avoiding the two-stage paradigm requirement of RAIR/RSR. Computational models as a class carry a shared limitation: they require non-trivial expertise to fit and interpret, which likely explains limited adoption.

4.4.2. Eye-Tracking

Cao et al. (2022) is the only corpus paper to deploy eye-tracking as a primary process measure, computing Percent Gaze Duration on the AI Suggestion alongside behavioral user-AI agreement and a self-report reliance item. The three measures showed moderate but imperfect alignment: participants who gazed more at the AI suggestion were more likely to follow it, but the correlation was far from perfect. A participant could attend closely to AI advice and reject it (appropriate under-reliance with full information) or accept it without attending to it (automatic reliance). Eye-tracking isolates the attentional component and should be combined with behavioral measures rather than used alone.

4.4.3. EEG and Physiological Measures

Zhang et al. (2024) is the sole paper combining neural data with behavioral reliance measures, adding EEG recording (32 electrodes) to RAIR/RSR in an image classification task (N=34, lab-based). A Mental Workload Index derived from frontal theta power partially mediated the XAI-to-RAIR pathway: explanations reduced workload in high-difficulty conditions, and reduced workload enabled more appropriate AI-following. The mediation provides a mechanistic account unavailable to behavioral studies alone, but the small sample and lab setting limit generalization.

4.4.4. Think-Aloud and Interaction Logs

Chen et al. (2023) used concurrent think-aloud protocols with grounded theory coding, identifying three distinct non-reliance pathways: outcome intuition (prior feeling about the correct answer), feature intuition (applying domain knowledge to evaluation highlights), and AI limitation intuition (recognizing error patterns). This taxonomy explains why the same XAI format can produce appropriate reliance through different mechanisms in different participants; a result behavioral outcomes alone cannot reveal. The approach is labour-intensive and susceptible to reactivity, but uniquely suited to mechanistic questions.

Mei et al. (2025) operationalized Time Spent Reading AI Explanations as a process measure alongside behavioral and self-report outcomes (N=810), finding explanation reading time positively associated with better-calibrated decisions. Lee and Chew (2023) found counterfactual explanations added ~18 seconds per decision relative to salient features alone; a temporal cost of the cognitive engagement required for more appropriate reliance. Response time captures effort but conflates reading, deliberation, and motor time, and cannot directly indicate whether additional time produced productive evaluation.

4.4.5. Measurement Properties

The process-level category is unified by mechanistic aspiration but fragmented across incompatible approaches; computational models, gaze, EEG, think-aloud, and interaction logs illuminate different facets of the reliance process but share no common metric. Ground truth is available in 9 of 11 papers (82%). Normative grounding is partial and instrument-dependent: measures paired with RAIR/RSR (Zhang et al., 2024) or fitted to correctness-conditioned outcomes (Tejeda et al., 2022) can evaluate whether reliance was warranted, whereas think-aloud coding and interaction logs index the process of reliance without evaluating its appropriateness. Separation of over- and under-reliance is similarly partial: cognitive models and eye-tracking can distinguish cases where the participant followed a correct AI from cases where they resisted a correct one (Cao et al., 2022; Li et al., 2023), but only when the underlying behavioral data are correctness-conditioned; without that conditioning, process measures index engagement intensity rather than reliance direction. The category's most consistent theoretical contribution is demonstrating that reliance is not a unitary act: attention and adoption (Cao et al., 2022), workload and accuracy (Zhang et al., 2024), verbalized reasoning and behavioral outcome (Chen et al., 2023), and engagement time and decision quality (Mei et al., 2025) can all dissociate. This implies that single-measure designs will systematically miss variance in the reliance construct, and that process-level instruments should be integrated into multi-method designs rather than treated as supplementary curiosities.

4.5. Multi-Method Approaches and Measure Co-occurrence

Of the 67 included papers, 41 (61%) use instruments from at least two categories simultaneously. This prevalence reflects researcher awareness that no single measure fully captures reliance, but the most important contribution of multi-method studies is not breadth but *divergence*: the majority of multi-method papers report findings where their measures produce different or incompatible signals.

4.5.1. *Patterns of Combination*

The most common pairing is behavioural and self-report (n=13), reflecting standard practice of collecting a trust scale alongside a behavioural outcome. Behavioural and performance-based measures co-occur in nine papers, typically pairing raw agreement rates with RAIR/RSR. Seven papers combine behavioral, performance-based, and self-report measures simultaneously (He et al., 2023; Salimzadeh et al., 2024; Küper et al., 2025), and two papers discuss all four measure types (Wischnewski et al., 2023; Lee & Chew, 2023). Process-level instruments are most often paired with behavioral measures (n=6 across two- and three-way combinations), consistent with their role as mechanistic supplements. The full co-occurrence structure is presented in Table 1.

Several papers explicitly integrate multiple measures into a single interpretive structure rather than simply collecting them in parallel. Mehrotra et al. (2024a) developed a formal classification scheme assigning each participant-trial to one of four categories (appropriate trust, over-trust, under-trust, inconsistency) by cross-referencing subjective trust, relative trustworthiness, and behavioral reliance simultaneously; a triadic integration that requires all three measure types to function. Kahr et al. (2024) tracked WOA and per-trial trust longitudinally across 20 sentencing cases, modeling their mutual influence via SEM. The key finding, that trust and WOA influence each other reciprocally but are not reducible to each other, provides the strongest within-paper evidence in the corpus that the two constructs are empirically as well as conceptually distinct.

4.5.2. *Measure Divergence as a Substantive Finding*

The most consistent divergence pattern is between self-reported trust and behavioral reliance, documented in at least 15 multi-method papers. Zhang et al. (2022) found trust tracked AI accuracy levels while behavioral agreement did not follow the same pattern. Scharowski et al. (2023) found no TPA differences across conditions despite meaningful WOA variation. Banovic et al. (2023) found participants rated the chess coaches they behaviorally relied on *least* most favourably on subjective ratings. Tolmeijer et al. (2021) found the reverse: self-reported trust was the more sensitive measure, tracking accuracy sequences that behavioral rates missed. The direction varies, but the dissociation is consistent: the two instruments measure different things.

A second pattern concerns the relationship between over-reliance and trust. Lee and Chew (2023) found participants preferred the salient-features system on trust and usability ratings, yet the counterfactual condition produced lower over-reliance rates and better team performance; a case where preferred system and better-calibrated system were opposite. Küper et al. (2025) found trust explained up to 24.1% of WOA variance but RAIR remained very low (10.04%) even among high-trust dermatologists, demonstrating that trust is necessary but not sufficient for appropriate AI-following in expert populations.

A third pattern concerns RAIR and RSR themselves. Multiple papers find they can move independently: Morrison et al. (2024) found explanations improved RAIR without changing RSR; He et al. (2023) found calibration tutorials selectively improved RAIR; Bo et al. (2025) found LLM explanation formats differentially affected RLR and RSR. These intra-framework dissociations establish that over-reliance and under-reliance are not opposite ends of a single dimension; they are partially independent tendencies that respond to different interventions, with direct implications for XAI design.

A fourth pattern concerns behavioural and process-level divergence. Cao et al. (2022) found attentional and behavioural reliance moderately but imperfectly correlated; participants sometimes attended closely to AI advice they did not adopt, and vice versa. Pitts et al. (2025) found participants' overreliance self-assessment systematically underestimated their actual behavioural rate, a metacognitive blind spot self-report alone could not have revealed.

4.5.3. Measurement Properties

Multi-method studies establish three findings invisible at the level of individual measure types. First, measure divergence is the norm rather than the exception: the majority of the 41 multi-method papers report at least one incompatible signal across their instruments. The choice of measure is therefore not a methodological convenience but a substantive decision about which facet of reliance a study is sensitive to. A study measuring only trust will miss the behavioural non-compliance documented by Klingbeil et al. (2024). A study measuring only agreement rate will miss the RAIR/RSR dissociation documented by Morrison et al. (2024). A study measuring only RAIR will miss the attentional dynamics and metacognitive blind spots documented by Cao et al. (2022) and Pitts et al. (2025). Second, the direction of self-report divergence is not random: self-reported trust consistently over-estimates behavioural reliance (Wang et al., 2024; Pitts et al., 2025; Scharowski et al., 2023), implying that reliance research conducted exclusively with self-report instruments will inflate estimates of AI adoption and underestimate the prevalence of under-reliance. Third, no paper uses more than four measure types simultaneously, and most multi-method papers pair only two - suggesting that even researchers who recognise measurement complexity are constrained by participant burden and analytic tractability from achieving genuinely comprehensive coverage. The full co-occurrence profile across all four categories is summarised in Table 1; the prescriptive framework in Section 6 is designed to help close the coverage gap.

5. Analysis and Critique

The taxonomy presented in Section 4 describes what the field does. This section evaluates whether what the field does is adequate for the construct it claims to measure. Five interconnected problems emerge from the corpus:

construct ambiguity, the systematic conflation of trust with reliance, measurement insensitivity to the dynamic and partial nature of reliance, a critical ecological validity gap, and a fragmentation of measurement approaches that prevents cumulative scientific progress. These problems are not independent, they reinforce each other, and together they suggest that the field's measurement practices have not kept pace with its theoretical ambitions. Table 2 summarises all five themes and their empirical evidence in the selected papers.

Table 2. Cross-Cutting Critiques of Reliance Measurement Practice: Prevalence, Evidence, and Implication

Critique	Prevalence in corpus	Key evidence	Implication
Construct ambiguity	16% of papers operationalise reliance without defining the construct; definitions vary substantially across the remaining 84%	Incompatible definitions in the same corpus: reliance as a trial-level behavioural act (<i>Schemmer et al., 2022</i>), a dispositional pattern (<i>Hou et al., 2025</i>), and a volitional engagement choice (<i>Zhou et al., 2025</i>)	A shared definitional standard is a prerequisite for instrument comparability; without it, measure selection reflects assumed rather than agreed construct boundaries
Trust–reliance conflation	30% of papers use trust calibration as reliance proxy; 67% of self-report papers do so	28/33 multi-method papers (85%) report trust–reliance divergence; self-report systematically over-estimates behavioural reliance (<i>Wang et al., 2024; Pitts et al., 2025</i>)	Trust calibration scores are not valid proxies for behavioural reliance; studies relying solely on trust scales cannot detect over- or under-reliance
Static measurement of a dynamic construct	62/67 papers (93%) report reliance as a cross-trial aggregate; only 1 genuinely longitudinal design in the corpus	<i>Li et al. (2023)</i> demonstrate asymmetric trust dynamics invisible to aggregate scores; <i>Kahr et al. (2024)</i> show reciprocal longitudinal trust–reliance influences that aggregation erases	Reliance trajectories, anchoring effects, and individual heterogeneity are structurally invisible to current measurement practice
Ecological validity gap	78% online/crowdsourced samples; median N = 169; 93% single-task designs; expert populations consistently show divergent reliance patterns	Dermatologists showed RAIR of 10.04% vs. near-chance lay rates (<i>Küper et al., 2025</i>); judges reject AI risk scores on professional identity grounds (<i>Pruss et al., 2023</i>)	Corpus findings may substantially misrepresent reliance in real deployed AI contexts; expert reliance operates through different mechanisms than lay reliance
Measurement fragmentation	15+ distinct reliance measures in the corpus; XAI effects not synthesizable across 21 papers using incompatible outcomes	Four non-identical RAIR/RSR implementations across 9 papers; <i>Chen et al. (2023)</i> and <i>Robbemond et al. (2022)</i> study the same XAI question with incomparable measures and incomparable conclusions	Cumulative science requires standardization of at least a reporting minimum; current findings cannot be integrated or meta-analysed

Note. Key evidence cells cite the most directly probative empirical finding for each critique. Full supporting evidence is developed in Section 5.

5.1. Construct Ambiguity: Measuring Before Defining

The most foundational problem in the sample is definitional. Eleven of the 67 included papers (16%) operationalise a reliance measure without providing a formal conceptual definition of the construct being measured (Leist et al., 2025; Papenmeier et al., 2022; Robbemonnd et al., 2022; Pruss et al., 2023; Ahn et al., 2024; Jong et al., 2025; Ramesh et al., 2020; Chen et al., 2025; Steyvers et al., 2024; Cao et al., 2022; Wang et al., 2024). Among the remaining 56 papers that do provide a definition, the definitions themselves are heterogeneous enough to cover fundamentally different phenomena: some frame reliance as the behavioural act of following an AI recommendation on a given trial (Schemmer et al., 2022; He et al., 2023), others as a disposition or tendency to defer to automated systems across contexts (Hou et al., 2025a; Hou et al., 2025b), others still as a calibration state in which trust is appropriately matched to system capability (Wischnewski et al., 2023; Naiseh et al., 2023), and a small group as a volitional choice to engage with AI output at all (Zhou et al., 2025; Mei et al., 2025). These are not minor terminological variations as they imply different measurement paradigms, different study designs, and different normative standards for what counts as appropriate reliance.

The consequence is that studies nominally measuring the same construct produce incommensurable results. A study using WOA to measure continuous adjustment toward AI advice (Holstein et al., 2025; Kahr et al., 2024) is not measuring the same thing as a study using RAIR to measure the proportion of trials where correct AI advice was followed (Schemmer et al., 2023), which is not the same as a study using the GenAI Reliance Scale to measure habitual dependence on generative AI tools (Hou et al., 2025a). Cross-study comparisons are therefore largely uninterpretable, and meta-analytic synthesis of effect sizes remains impossible without a shared definitional foundation. The field's most urgent theoretical task is not the development of new measures but the consolidation of a shared construct definition from which measurement can proceed.

5.2. The Trust–Reliance Conflation

The second and perhaps most persistent problem is the systematic substitution of trust calibration for reliance measurement. Twenty of the 67 included papers (30%) treat trust calibration, the alignment between self-reported trust and actual system trustworthiness, as a proxy for behavioural reliance, without collecting a direct reliance outcome. This conflation is most prevalent in the self-report category, where 16 of 24 papers (67%) use trust scales as their primary outcome, but it also appears in 8 of 40 behavioural papers and 5 of 24 performance-based papers. The theoretical basis for this substitution, that well-calibrated trust produces well-calibrated reliance behaviour, was articulated by Lee and See (2004) and has been influential. But the empirical evidence within the sample directly challenges its validity. Twenty-eight of the 33 multi-method papers (85%) that collect both trust and reliance measures report at least one finding where the two diverge in direction or magnitude. The

pattern of divergence is not random: self-report trust measures almost systematically over-estimate reliance. Wang et al. (2024) found that per-trial self-reported reliance was consistently higher than behavioural agreement rates across all experimental conditions. Pitts et al. (2025) found that participants systematically underestimated their own behavioural over-reliance. Banovic et al. (2023) found that participants rated chess coaches most highly on subjective scales precisely when they relied on them least behaviourally. Scharowski et al. (2023) found no significant difference in TPA trust scores between conditions despite meaningful WOA variation, concluding the two instruments measure distinct constructs. Lee and Chew (2023) found that participants trusted the salient-features condition more while achieving better-calibrated behavioural reliance under counterfactual explanations which is a direct inversion of the assumed trust-reliance correspondence. Table 4 shows the direction of divergence across the key multi-method papers in the corpus.

The problem is compounded by terminological slippage throughout the sample. Several papers use the phrase "behavioural trust" to label switching rates (Papenmeier et al., 2022) and "trust calibration" to describe what is functionally a reliance calibration paradigm (Steyvers et al., 2024; Mehrotra et al., 2024a), making it difficult for readers to identify which construct is actually being measured. The implication is unambiguous: trust and reliance should be measured separately, treated as conceptually and empirically distinct constructs, and their relationship treated as an empirical question rather than a methodological shortcut.

5.3. Measurement Insensitivity: Static Snapshots of a Dynamic Construct

The third problem is that the dominant measurement paradigms treat reliance as a stable cross-trial aggregate when the theoretical and empirical evidence shows it is neither stable nor uniform across individuals. Reliance evolves across trials as participants update their mental model of the AI system, receive feedback on prior decisions, and respond to anchoring effects from early encounters (Lee & See, 2004; Nourani et al., 2021). Yet 62 of the 67 papers in the corpus reduce this trajectory to a single mean score, discarding all temporal information.

The few papers that do track reliance dynamically reveal how much is lost by aggregation. Li et al. (2023) demonstrated using a Hidden Markov Model that trust state transitions are asymmetric, negative experiences degrade reliance more rapidly than positive experiences restore it, a finding entirely obscured by aggregate measures. Tolmeijer et al. (2021) found that the sequence of correct and incorrect AI outputs mattered as much as their overall frequency in shaping behavioural reliance trajectories. Kahr et al. (2024) modelled WOA and self-reported trust longitudinally across 20 trials, finding reciprocal influence between the two constructs that was invisible in cross-sectional designs. Nourani et al. (2021) showed that early encounters with AI errors produced anchoring effects that persisted even after participants had observed equal numbers of correct outputs, a primacy bias that trial-averaged measures cannot detect. Spatola et al. (2024) found that reliance inertia, the tendency to

maintain established reliance patterns even when AI reliability changes, transferred across tasks, suggesting that reliance history is itself a substantively important variable the field currently treats as noise.

The sensitivity problem extends to individual heterogeneity. Most studies report group-level means that aggregate across participants with very different reliance profiles. Lu et al. (2024) showed using Bayesian mixture modeling that participant populations contain distinct latent reliance types, conservative, neutral, and liberal AI-followers, entirely obscured by group means. Holstein et al. (2025) found that median WOA collapsed to zero at very high uncertainty levels while mean WOA remained positive, a distributional property lost in aggregate reporting. A field that reports only means cannot identify the sub-populations most at risk of over-reliance, which is arguably the most practically important question in human-AI safety research.

5.4. The Ecological Validity Gap

The fourth problem is the severe mismatch between the conditions under which reliance is studied and the conditions under which AI systems are deployed. Fifty-two of the 67 papers (78%) rely on online crowdsourced samples recruited through Prolific or Amazon Mechanical Turk, with a median sample size of 169. Ten papers use fewer than 50 participants (insufficient power for the multi-condition designs many of them employ). Sixty-two of 67 papers use a single task in a single domain. Only one paper tracks reliance over an extended period in a naturalistic deployment setting.

Domain homogeneity is a particular concern. The corpus is dominated by general or abstract classification tasks ($n=20$) and hypothetical healthcare and legal scenarios, with only a small number of studies using real expert populations or genuine deployment contexts. The handful of studies that do use expert participants consistently find reliance patterns that diverge sharply from lay samples. Küper et al. (2025) found that 223 dermatologists showed extremely low RAIR (10.04%) and very high RSR (85.6%), strong under-reliance, in a pattern that would be invisible in a crowdsourced sample. Goddard et al. (2014) found that clinical experience strongly and negatively predicted switching among GPs, with experienced clinicians substantially less prone to over-reliance than novices. Pruss et al. (2023) found that judges systematically did not engage with a deployed AI sentencing tool, with organizational and professional-identity factors, not trust or reliance calibration, as the primary explanatory variables. Cabitza et al. (2025) found automation bias in spine surgeons and radiologists but at substantially lower rates than typically reported in crowdsourced studies. These expert-population findings are not outliers, they are the ecologically valid signal that the crowdsourced corpus is systematically unable to detect.

A related validity threat concerns AI accuracy levels. Most studies fix AI accuracy between 65% and 90% with a precisely controlled proportion of correct and incorrect trials (Schemmer et al., 2023; He et al., 2023; Li et al., 2025; Naiseh et al., 2023). Real deployed systems do not operate at fixed accuracy levels, do not present

errors in controlled proportions, and do not signal their errors to users through the study design. Reliance patterns observed under artificially controlled accuracy may not generalize to deployment environments where the AI's reliability is itself unknown and variable.

5.5. Measurement Fragmentation and the Absence of Cumulative Science

The fifth problem is that the field's measurement practices prevent the accumulation of comparable and replicable findings. The selected papers contain at least fifteen distinguishable reliance measures, and very few papers use the same measure as the study they most directly build on. Even within the most standardised sub-category, RAIR/RSR, there are at least four non-identical implementations across the nine papers that use the framework (Schemmer et al., 2022; Schemmer et al., 2023; He et al., 2023; Bo et al., 2025), differing in how the denominator is defined, how ties are handled, and whether the two measures are reported separately or collapsed into a scalar. Papers studying nominally identical interventions use different tasks, different AI accuracy levels, different measure sub-types, and different populations, making cross-study synthesis impossible.

The XAI literature illustrates the consequences most clearly. Twenty-one papers in the corpus examine explanation effects on reliance, and of these the majority report null or negative findings, i.e., explanations do not improve, or actively worsen, behavioural reliance calibration (Chen et al., 2023; He et al., 2024; Naiseh et al., 2022; Doh et al., 2025; Li et al., 2025). The apparent contradiction between the theoretical promise of XAI and its empirical reliance effects has been noted by individual papers (Buçinca et al., 2021; Chen et al., 2023; He et al., 2023) but cannot be resolved at the field level because the studies use incompatible outcome measures. A study finding that feature-based explanations improve accuracy (Robbemon et al., 2022) is not directly comparable to a study finding the same explanation type reduces RAIR (He et al., 2024) or increases WOA without improving appropriateness (Ahn et al., 2024). The field is generating a large body of findings that cannot be integrated because the measurement infrastructure does not support integration.

6. Discussion

The preceding sections have mapped the measurement landscape of AI reliance research in HCI, identified the structural properties of the five major measurement categories, and diagnosed five cross-cutting problems that limit the field's capacity to accumulate comparable and actionable findings. This section translates those findings into a prescriptive direction. We first propose a measurement framework, a set of decision rules and minimum standards to guide measure selection and reporting in future reliance research. We then draw out the broader theoretical implications of the corpus for how reliance is conceptualized in human-AI interaction. We close with an honest account of the review's limitations and the most productive directions for future empirical work.

6.1.A Prescriptive Measurement Framework

The central practical contribution of this review is the observation that the choice of reliance measure is not a neutral methodological convenience, instead it is a substantive decision that determines which aspect of the reliance construct a study is sensitive to, and therefore which findings it can produce. A study using only a self-reported trust scale cannot detect the behavioural non-compliance (Klingbeil et al. 2024; Banovic et al. 2023). A study using only an agreement rate cannot detect the RAIR/RSR dissociation (Morrison et al. 2024 ;He et al. 2023). A study using only a mean WOA cannot detect the distributional heterogeneity (Holstein et al. 2025) or the temporal anchoring (Nourani et al. 2021). The field does not need a single universal measure, the right measure depends on the research question, study design, and deployment context, but it does need a principled framework for selecting among measures and a common reporting standard that applies regardless of which measure is chosen.

We propose such a framework organized around two components: a measure selection guide keyed to research question type, and a minimum reporting standard applicable to all reliance studies. Table 2 presents the measure selection guide.

Table 3. *Prescriptive Measure Selection Guide for Human-AI Reliance Research*

Research question type	Recommended primary measure	Required design elements	Acceptable alternatives	Measures / designs to avoid
Is reliance appropriate? (characterising over- vs. under-reliance in a sample)	RAIR + RSR (<i>Schemmer et al., 2022</i>)	Two-stage design; ground truth; ≥50% AI-incorrect trials	Over/under-reliance rate (<i>Li et al., 2025</i>); HASC matrix (<i>Doh et al., 2025</i>)	Raw agreement rate alone; trust scale alone
How much do participants adjust toward AI advice ?	Weight on Advice (WOA) (<i>Yaniv & Kleinberger, 2000</i>)	Continuous response scale; two-stage design; ground truth optional	Agreement & Switch Fraction (<i>He et al., 2023</i>)	Binary accept/reject rate alone
Does an intervention change reliance direction or magnitude ? (experimental design)	RAIR + RSR (binary tasks) or WOA (continuous tasks)	Two-stage design; ground truth; pre-registered AI accuracy; manipulation check	Over/under-reliance rate; Agreement & Switch Fraction	Agreement rate or accuracy as sole outcome; no pre-AI baseline
What is participants' habitual or dispositional reliance on AI ?	GenAI Reliance Scale (<i>Hou et al., 2025a</i>)	Survey design; no trial-level task required; test-retest for reliability	Custom single-item reliance rating with test-retest validation	Jian or TiA trust scales as reliance proxy
How does reliance evolve across trials or sessions ?	Trial-by-trial switch rate or WOA with trajectory modeling (<i>Li et al., 2023</i> ; <i>Lu et al., 2024</i>)	≥20 trials; repeated measures; feedback condition specified; individual-level analysis	HMM (<i>Li et al., 2023</i>); Bayesian mixture model (<i>Lu et al., 2024</i>); Kahr et al. SEM approach (2024)	Single aggregate score across all trials

Research question type	Recommended primary measure	Required design elements	Acceptable alternatives	Measures / designs to avoid
What mechanisms underlie reliance decisions?	Computational cognitive model (<i>Li et al., 2023; Lu et al., 2024</i>) + behavioral measure	Sufficient trials for model fitting; individual-level behavioral data; ground truth	Eye-tracking (<i>Cao et al., 2022</i>); EEG (<i>Zhang et al., 2024</i>); think-aloud (<i>Chen et al., 2023</i>); interaction logs (<i>Mei et al., 2025</i>)	Self-report alone; single aggregate behavioral score
How does reliance differ across expert and lay populations?	RAIR + RSR with domain-specific stimuli (<i>Küper et al., 2025; Goddard et al., 2014</i>)	Expert recruitment; domain-valid task; ground truth from domain standard; matched lay comparison group	WOA with continuous scale appropriate to domain; qualitative non-use coding	Crowdsourced sample as expert proxy; generic classification task with no domain validity

Note. "Measures / designs to avoid" refers to using the listed instrument or approach as the sole measure for the stated research question, not to its use as a supplementary parallel measure. Rows 1 and 3 both recommend RAIR + RSR but are distinguished by purpose: Row 1 characterises an existing sample's reliance profile; Row 3 tests whether a manipulation changes it, requiring additional design controls (pre-registration, manipulation check). All studies should additionally satisfy the Minimum Reporting Standard (Section 6.1).

Beyond measure selection, we propose a minimum reporting standard (a set of items that every reliance study should report regardless of which primary measure it uses). These items address the recurrent sources of incommensurability identified in Section 5 and are designed to be low-burden: most can be reported in one or two sentences in the methods section.

Minimum Reporting Standard for Reliance Studies:

1. *Construct definition.* A brief explicit statement of how reliance is defined in the study, behavioural act, disposition, calibration state, or engagement choice, and the citation from which the definition is drawn or adapted.
2. *Measure justification.* A statement linking the chosen measure to the research question, including acknowledgment of what the measure cannot capture (e.g., "WOA does not distinguish appropriate from inappropriate reliance without ground truth conditioning").
3. *Ground truth status.* Whether a ground truth was available for each trial, how it was determined, and whether it was revealed to participants.
4. *AI accuracy and error distribution.* The proportion of AI-correct and AI-incorrect trials, how this was determined or controlled, and whether it was fixed or variable across the study.
5. *Trust measurement status.* Whether trust was measured, which instrument was used, and whether trust is being used as a reliance proxy or as a conceptually distinct parallel measure.
6. *Temporal structure.* Whether reliance is reported as a cross-trial aggregate or with trial-level resolution, and if aggregate, the number of trials over which it is computed.

7. *Sample characteristics.* Participant expertise level, recruitment platform, and any prior experience with similar AI systems.

These seven items do not constrain research design, they make research designs interpretable across studies. If adopted as a reporting norm by HCI venues, they would substantially reduce the incommensurability problem identified in Section 5.5 without requiring any convergence on a single measure.

6.2. Theoretical Implications

Beyond the prescriptive framework, the sample papers generate three theoretical implications that go beyond measurement methodology and speak to how reliance is conceptualised in human-AI interaction research.

The first and most consequential is that improving trust calibration is not sufficient to improve reliance calibration. The dominant theoretical assumption in the XAI literature, that better-calibrated trust, achieved through more informative explanations, will produce better-calibrated reliance behaviour, is challenged by the systematic trust–reliance dissociation documented in Section 5.2. If 85% of multi-method studies find that trust and reliance measures diverge, and if the direction of divergence is consistently toward trust over-estimating reliance, then trust calibration is not a reliable leading indicator of reliance calibration. This has a direct implication for XAI evaluation: XAI systems should not be evaluated primarily on their effects on self-reported trust or trust calibration. They should be evaluated on their effects on behavioural reliance, specifically on RAIR and RSR separately, because the evidence shows that interventions can improve one without improving the other (Morrison et al., 2024; He et al., 2023). A field that continues to evaluate XAI against trust outcomes while expecting reliance benefits will continue to produce the pattern of XAI null findings documented in Section 5.5.

The second theoretical implication concerns the temporal structure of reliance. The field currently treats reliance as a cross-sectional state, a property of a person-AI-task triad at a given moment, when the dynamic evidence suggests it is better understood as a trajectory shaped by cumulative experience, anchoring effects, and inertia. Li et al.'s (2023) asymmetric trust update model, Kahr et al.'s (2024) longitudinal reciprocal influence model, and Spatola et al.'s (2024) transfer-of- reliance finding all point toward the same conclusion: reliance at trial t is a function of reliance at trial $t-1$, and any study that does not model this dependency is treating a temporal process as a static state. The theoretical implication is that the field needs a process model of reliance formation, an account of how reliance is initialized, updated, and transferred across contexts, rather than just a taxonomy of reliance states. Computational cognitive modelling approaches such as those used by Li et al. (2023) and Lu et al. (2024) are the most promising methodological template for developing such a model, and their current under-use (2 papers out of 67) represents a significant gap.

The third theoretical implication concerns the role of expertise. The expert-population evidence reviewed in Section 5.4 does not merely demonstrate that experts and lay people differ in their reliance rates, it suggests that the mechanisms driving reliance may be qualitatively different across expertise levels. In lay samples, over-reliance is the dominant failure mode and trust is a reliable predictor of behavioural following. In expert samples, under-reliance is common, professional identity and domain experience are the primary moderators, and trust scales are less predictive (Küper et al., 2025; Goddard et al., 2014; Pruss et al., 2023). This suggests that the theoretical framework most commonly applied in the corpus, in which reliance is a function of perceived AI trustworthiness and task difficulty, may be a good model of novice reliance but a poor model of expert reliance. Expert reliance may be better modelled as a function of AI output consistency with professional schema, organizational norms around tool use, and the epistemic cost of overriding AI output in accountable professional settings. HCI research that aims to support appropriate reliance in high-stakes deployment contexts, clinical decision support, legal analytics, financial advisory, will require a theoretically richer account of expert reliance than the current literature provides.

6.3. Limitations and Future Directions

This review has several limitations that bound the interpretation of its findings. The corpus was constructed using a search strategy cantered on HCI and human-AI decision-making literature. This strategy prioritised depth and coding quality over exhaustive breadth. As a result, research on human-AI reliance published primarily in cognitive psychology (Dietvorst et al., 2015), organizational behaviour (Logg et al., 2019), judgment and decision-making (Yaniv & Kleinberger, 2000), and natural language processing venues is underrepresented. The taxonomy and critique therefore reflect the HCI field's measurement practices specifically rather than the broader multidisciplinary literature on human-AI decision-making. Future reviews should explicitly integrate these adjacent literatures, as they contain both more mature measurement traditions, particularly in the judge-advisor paradigm, where WOA and related measures have decades of validation, and more ecologically valid study designs, particularly in organizational and field research.

The review is also limited to English-language publications, which may underrepresent reliance research conducted in non-Western deployment contexts where human-AI interaction norms, professional cultures, and regulatory environments differ substantially. The AI accuracy levels, task domains, and participant populations that dominate the corpus reflect the research infrastructure of North American and European HCI communities, and the generalizability of the taxonomy to other research contexts should be treated as an open question. Additionally, the corpus spans 2014 to 2026, a period of rapid change in AI capability and deployment; reliance dynamics observed with earlier supervised ML systems may differ from those emerging with large language

models, and the GenAI-specific papers in the corpus (Hou et al., 2025a; Jong et al., 2025; Kim et al., 2025) suggest that measurement approaches developed for structured classification tasks may require adaptation for open-ended generative AI interactions.

Within these boundaries, the selected points to four productive directions for future empirical work. First, longitudinal field studies tracking reliance over weeks or months in real deployment settings, clinical, legal, or financial, are the most significant gap in the current literature. Such studies would address simultaneously the ecological validity problem, the static measurement problem, and the expert population problem, and would provide the data needed to validate or refute the dynamic reliance process models proposed by Li et al. (2023) and Kahr et al. (2024). Second, cross-domain reliance generalisability studies, in which the same participants complete reliance paradigms in two or more task domains, would establish whether reliance is a stable individual difference trait or a domain-specific tendency, a question with important implications for both theory and system design. Third, standardization efforts within the RAIR/RSR family of measures, agreeing on denominator definitions, handling of ties, and reporting conventions, would immediately improve the synthesizability of the most promising performance-based measurement tradition without requiring new data collection. Fourth, computational modelling of individual-level reliance trajectories, building on the HMM and Bayesian mixture approaches in the corpus, offers the most methodologically powerful path toward understanding the mechanisms of reliance formation and the identification of at-risk sub-populations.

7. Conclusion

This review began from a simple observation: the field studying human reliance on AI has invested heavily in empirical production without investing comparably in measurement scrutiny. The instruments through which reliance is studied determine what findings are possible, and those instruments had never been systematically examined. We have conducted that examination across 67 papers published between 2014 and 2026, and the picture that emerges is one of genuine methodological diversity deployed without sufficient discipline. Different measures are appropriate for different questions, but that only holds if researchers select instruments with an understanding of what each cannot capture, and the selected papers show they routinely do not. The consequence is that trust is substituted for reliance, reliance is aggregated across trials despite being a temporal trajectory, crowdsourced samples stand in for expert populations, and fifteen-plus distinguishable measures are deployed without a shared reporting standard. These are not minor imperfections; they are structural barriers to cumulative science that the prescriptive framework in Section 6 is designed to lower. Reliance cannot be treated as a stable individual attribute to be measured once and summarised as a score. It is a dynamic behavioural pattern shaped by AI accuracy experience, expertise, professional context, and the structure of the decision

environment. Whether the field's rapidly growing empirical base becomes the foundation for a cumulative science of human-AI calibration or remains a collection of individually interesting but irreconcilable studies, depends in large part on whether it takes its measurement practices seriously.

Competing Interests

The authors declare that they have no competing financial or personal interests that could have appeared to influence the work reported in this paper.

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Author Contribution

MN : Conceptualisation, Methodology, Data Collection, Analysis and Writing Original Draft

DC: Data Collection, Analysis, Review and Editing

HD: Data Analysis Validation, Review and Editing

NJ: Review and Editing

Data Availability Statement:

The full coded dataset can be found at the following link: (https://osf.io/f5bt9/overview?view_only=a3627a2301cc42cbb44dd0d5f1e99297 }

Research Involving Human and/or Animals

Not Applicable

Informed Consent

Not Applicable

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